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# Actual Flip-Angle Imaging in the Pulsed Steady State: A Method for Rapid Three-Dimensional Mapping of the Transmitted Radiofrequency Field

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**A new method has been developed for fast image-based measurements of the transmitted radiofrequency (RF) field. The method employs an actual flip-angle imaging (AFI) pulse sequence that consists of two identical RF pulses followed by two delays of different duration ($TR_1 < TR_2$). After each pulse, a gradient-echo (GRE) signal is acquired. It has been shown theoretically and experimentally that if delays $TR_1$ and $TR_2$ are sufficiently short and the transverse magnetization is completely spoiled, the ratio $r = S_2/S_1$ of signal intensities $S_1$ and $S_2$, acquired at the beginning of the time intervals $TR_1$ and $TR_2$, depends on the flip angle (FA) of applied pulses as $r = (1 + n * \cos(FA))/(n + \cos(FA))$, where $n = TR_2/TR_1$. The method allows fast 3D implementation and provides accurate $B_1$ measurements that are highly insensitive to $T_1$. The unique feature of the AFI method is that it uses a pulsed steady-state signal acquisition. This overcomes the limitation of previous methods that required long relaxation delays between sequence repetitions. The method has been shown to be useful for time-efficient whole-body $B_1$ mapping and correction of $T_1$ maps obtained using a variable FA technique in the presence of nonuniform RF excitation. Magn Reson Med 57:192–200, 2007. © 2006 Wiley-Liss, Inc.**

**Key words: $B_1$ mapping; radiofrequency field; $T_1$ mapping; flip-angle correction; whole body**

Knowledge of the spatial distribution of the transmitted radiofrequency (RF) field is needed for a number of MR research and engineering applications, such as correcting the results of various quantitative methods, validating theoretical models for electromagnetic field calculations, ensuring quality control of RF coils, and testing the MR compatibility of implanted objects, especially at high magnetic fields. While the RF field distribution can be obtained with the use of a small pickup coil or phantom measurements, direct in vivo $B_1$ mapping in the human body is always preferable, particularly for correcting patient-specific quantitative data, where individual variations in the $B_1$ field generally depend on the coil position, the dielectric properties of the tissues, and the subject's size. Although a number of methods have been proposed for image-based RF field measurements (1–17), many of them are difficult or impossible to use in vivo because they require serial scans at a variable transmitter power or pulse duration (1–7), have a high sensitivity to static magnetic field inhomogeneity (8,9), and result in high RF power deposition (1,3). These difficulties have been overcome by several $B_1$ mapping techniques that are more suitable for in vivo applications (10–17); however, further improvements in time efficiency, anatomical coverage, and accuracy are needed to introduce $B_1$ mapping into routine practice.

A common method for $B_1$ mapping is based on two scans acquired with a spin-echo (SE) or gradient-echo (GRE) imaging sequence at two nominal flip angles (FAs, usually $\alpha$ and $2\alpha$) chosen in a range of optimal sensitivities to $B_1$ variations for a particular sequence (10). To avoid the dependence of the signal on $T_1$, the repetition time (TR) of the sequence should be sufficiently long to achieve complete relaxation of magnetization (10). More time-efficient variants of the double-angle method (11–15) include driven recovery (11) or presaturation (15) of magnetization to reduce the TR, and its combination with fast acquisition sequences, such as fast SE (FSE) (12,13), echo-planar (14), and spiral (15). As an alternative to the double-angle approach, the RF distribution can be determined from a single scan with the use of specially designed pulse sequences that simultaneously acquire two signals characterized by different dependencies on the FA (16,17). A general advantage of such methods for in vivo measurements is the absence of possible misregistration between the images used to calculate the FA distribution. One of these techniques (16) observes stimulated echo and SE signals generated by the three-pulse sequence with FAs $\alpha$–$2\alpha$-$\alpha$, where the ratio of signals depends on the FA as $\cos(\alpha)$. Another method (17) employs a two-pulse sequence with identical pulses that generate two free induction decay (FID) signals in rapid succession, which scan the equilibrium and the subsequent state of the longitudinal magnetization.

A general limitation of the above techniques is that their in vivo applications necessitate 2D slice-based acquisition due to scan-time constraints and the relatively long sequence TRs required to achieve full or at least substantial relaxation of the longitudinal magnetization. However, slice-based techniques present the problem of nonuniform excitation across the slice profile, which precludes the use of FAs derived from analytical signal equations as a direct measure of the $B_1$ field (5,7,13,17). Although this circumstance is often ignored, the dependence of the excitation profile on the FA may introduce significant errors in $B_1$ measurements that rely on simple theoretical models derived for ideal pulses, especially if slice excitation is applied at different nominal FAs, such as in the double-angle (10–15) or stimulated-echo (16) methods. An alternative approach that utilizes nonselective RF pulses (12) is feasible only for single-slice acquisition, and correspondingly

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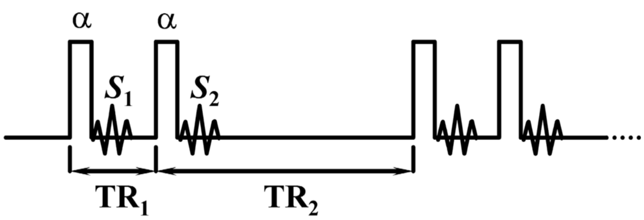


FIG. 1. Timing diagram of the AFI pulse sequence.

results in reduced spatial coverage or increased scan time. A common technique for correcting the slice-profile effect is based on computed excitation profiles, which depend on the pulse shape and FA (7,13,17). This approach adds complexity to the data processing and does not take into account an actual excitation profile, which may differ from the theoretical one due to the imperfectness of applied gradient pulses and variability of relaxation properties in tissues. A 3D acquisition technique that inherently produces rectangular slice profiles (5) would provide a natural solution for the excitation profile problem in $B_1$ mapping if a practically reasonable scan time could be achieved. This study presents a detailed description of a recently reported actual flip-angle imaging (AFI) method (18). This technique enables fast 3D acquisition of the actual FA distribution in the entire object while it also satisfies other requirements for in vivo measurements, such as $T_1$ insensitivity and a single-scan procedure. As an alternative to previous methods, AFI can be used to obtain RF field measurements with a much faster repetition rate of the sequence, resulting in a pulsed state of magnetization, and requires no substantial relaxation.

## THEORY

Consider a pulse sequence (referred to below as the AFI sequence) that consists of two identical RF pulses with the FA α followed by two delays $TR_1$ and $TR_2$ (Fig. 1). The FID signals $S_1$ and $S_2$ are observed after the corresponding pulses in a GRE form. Of note, a similar sequence was proposed for $B_1$ measurements in a previous study (17), which presented an analysis of the fully relaxed condition, in which the magnetization after each period $TR_2$ returns to the equilibrium state. With the additional assumption that longitudinal relaxation is negligible during $TR_1 \ll T_1$, in Ref. 17 the method was used to measure the actual FA from the signal ratio $r = S_2/S_1 = \cos\alpha$. Here we assume a more general situation, in which the relaxation effect is non-negligible, and a fast repetition rate of the sequence (i.e., $TR_1 < TR_2 < T_1$) results in establishing a pulsed steady state of magnetization. We further assume that the sequence is ideally spoiled, i.e., all transverse coherencies are irreversibly destroyed at the end of the delays $TR_1$ and $TR_2$. In the pulsed steady state, a consecutive solution of the Bloch equations for the AFI sequence results in expressions for the longitudinal magnetization before each excitation pulse:

$$M_{z1} = M_0 \frac{1 - E_2 + (1 - E_1)E_2 \cos\alpha}{1 - E_1 E_2 \cos^2\alpha}, \qquad [1]$$

$$M_{z2} = M_0 \frac{1 - E_1 + (1 - E_2)E_1 \cos\alpha}{1 - E_1 E_2 \cos^2\alpha}, \qquad [2]$$

where $E_{1,2} = \exp(-TR_{1,2}/T_1)$, and $M_0$ is the equilibrium magnetization. The observed signals $S_1$ and $S_2$ are proportional to the above magnetizations:

$$S_{1,2} = M_{z1,2} \exp(-TE/T_2^*)\sin\alpha, \qquad [3]$$

and their ratio can be expressed as:

$$r = S_2/S_1 = \frac{1 - E_1 + (1 - E_2)\, E_1 \cos\alpha}{1 - E_2 + (1 - E_1)E_2 \cos\alpha}. \qquad [4]$$

For short $TR_1$ and $TR_2$, Eq. [4] can be simplified by applying the first-order approximation to exponential terms:

$$r \approx \frac{1 + n\cos\alpha}{n + \cos\alpha}, \qquad [5]$$

where $n = TR_2/TR_1$. Thus, the ratio $r$ can be used as an independent of $T_1$ measure of the actual FA:

$$\alpha \approx \arccos \frac{rn - 1}{n - r}. \qquad [6]$$

For practical applications of the AFI method, an approximated Eq. [5] should provide sufficient accuracy for a variety of relaxation times $T_1$ observed in tissues, and the ratio $r$ must be sensitive to variations of the FA. Basic trends explaining the effects of pulse sequence parameters and relaxation times on the ratio $r$ are illustrated in Fig. 2, where simulated plots of the ratio $r$ are presented as functions of the FA for various relaxation times $T_1$ and timing parameters $TR_1$ and $TR_2$. Following Eq. [5], it is convenient to describe variations of $TR_2$ by its ratio to $TR_1$, $n = TR_2/TR_1$ at constant $TR_1$. Generally, the first-order approximation holds well for a wide variety of sequence parameters and $T_1$. Thus, to a high degree of approximation, $r$ turns out to be $T_1$-insensitive. A slight deviation from the first-order behavior occurs at high FAs for short $T_1$ (Fig. 2a). The effect of timing parameters on this deviation is minor, although an increase of $TR_1$ highlights disagreement between the exact (Eq. [4]) and first-order (Eq. [5]) solutions (Fig. 2b). At the same time, the ratio $r$ is practically unaffected by the $TR_1$ within a technically reasonable range of 10–30 ms, which allows all necessary gradient pulses (including relatively long spoiler gradients) in the sequence design to be accommodated for a variety of MR systems. The ratio $n$ appears to be the main parameter that determines the sensitivity of the AFI method to variations of the FA (Fig. 2c). An increase of $n$ amplifies the variability of $r$ that is equivalent to an improvement of the dynamic range of FA measurements. However, the parameter $n$ cannot be increased without limit due to scan time constraints. Based on simulation results, the choice of $n$ in a range of 4–6 provides sufficient sensitivity to FA variations, as reflected by a 15–65% decrease of the signal $S_2$ relative to $S_1$, if the FA varies in a range of 40–80° (Fig.

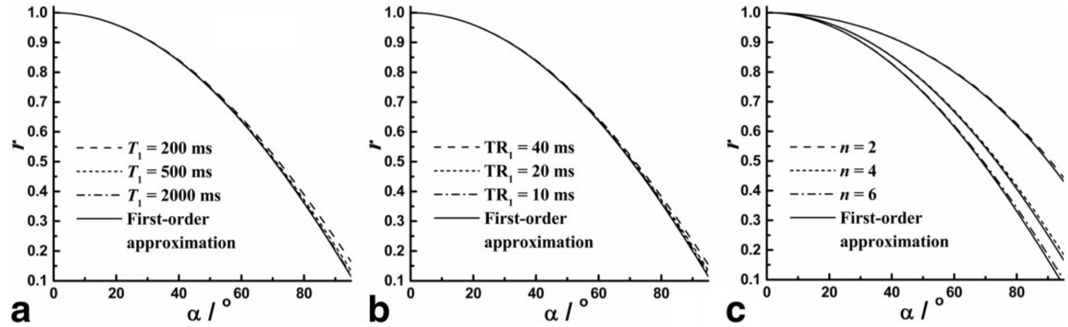


FIG. 2. Simulations of the signal ratio $r$ as a function of the FA for the AFI sequence. **a:** Effect of $T_1$. The plots were calculated by Eq. [4] for variable $T_1$ = 200 ms (dash line), 500 ms (short-dash line), and 2000 ms (dash-dot line) at $TR_1/TR_2$ = 20/100 ms (i.e., $n$ = 5). The first-order approximation (Eq. [5]) is plotted by the solid line. **b:** Effect of $TR_1$. The plots were calculated by Eq. [4] for $T_1$ = 500 ms at variable $TR_1/TR_2$ = 10/50 ms (dash-dot line), 20/100 ms (short-dash line), and 40/200 ms (dash line) corresponding to the fixed ratio $n$ = 5. The first-order approximation (Eq. [5]) is plotted by the solid line. **c:** Effect of $n$. The plots were calculated by Eq. [4] for $T_1$ = 500 ms and $TR_1$ = 20 ms at variable $n$ = 2 (i.e., $TR_2$ = 40 ms, dash line), 4 (i.e., $TR_2$ = 80 ms, short-dash line), and 6 (i.e., $TR_2$ = 120 ms, dash-dot line). The plots of the first-order approximation (Eq. [5]) are presented by solid lines that run close to the exact solutions for each $n$.

2c). For experimental $B_1$ measurements using the AFI sequence, this range of FAs is nearly optimal because at lower values the sensitivity is reduced, while at larger FAs the results can be influenced by the relaxation effect. At the above parameter settings, systematic errors caused by relaxation in a physiological $T_1$ range are expected to be negligible.

Finally, it should be noted that the limit of Eq. [5] at a large $n \gg \cos(\alpha)$ is $r \approx \cos(\alpha)$. Thus, the described method approaches the previous technique (17), which relies on the fully relaxed magnetization and negligible relaxation between pulses. However, the approach in Ref. 17 is inapplicable to the pulsed steady state, where both $TR_1$ and $TR_2$ are shorter than $T_1$, because the error of calculating $\alpha$ as $\alpha = \arccos(r)$ rapidly increases with a decrease of $n$. For example, even with a large $n$ = 10, this approximation underestimates the actual FA by 8–9% in a range of 40–80° for the model parameters $TR_1$ = 20 ms and $T_1$ = 500 ms.

## MATERIALS AND METHODS

The AFI sequence was implemented on a 1.5 T whole-body MR scanner (Signa Horizon EchoSpeed 5.8, General Electric, Milwaukee, WI, USA). The sequence enabled 3D signal acquisition and was equipped with the choice of rectangular nonselective or slab-selective RF pulses taken from the manufacturer's waveform library. The rectangular pulse was used with a duration of 0.4 ms. The slab-selective pulse had a minimum-phase waveform designed according to the Shinnar-Le Roux (SLR) algorithm (19) with a duration of 3.2 ms and a nominal bandwidth of 3664 Hz. Unless otherwise stated, the configuration of the sequence with the nonselective rectangular pulses was used in the experiments described below. A standard RF spoiling scheme with a 117° phase increment (20) was applied consecutively to each RF pulse. The sequence was implemented with eight dummy repetitions prior to data acquisition, and linear phase encoding that provided sufficient time (>5 $T_1$) for any physiologically reasonable $T_1$ to achieve the steady state when the central portion of the 3D $k$-space data was acquired. The method was validated in three phantom experiments and tested for $B_1$ mapping in the human body and head in vivo.

The first phantom experiment was intended to verify the theoretical concept of AFI for a series of known FAs. The experimental setup was designed to make nominal FAs prescribed at the scanner's console as close as possible to actual FAs produced by the sequence. A small phantom (approximate cross-sectional dimensions = 10 × 5 cm) containing media with different relaxation properties was placed in the center of the birdcage quadrature transmit-receive head coil (General Electric, Milwaukee, WI, USA). The phantom consisted of five tubes with 1.0, 0.5, and 0.1 mM solutions of gadolinium ($T_1$ = 202, 396, and 744 ms, respectively) prepared by diluting a commercial contrast agent (Omniscan; Nycomed, Princeton, NJ, USA), an approximately 0.3-mM solution of manganese chloride ($T_1$ = 426 ms), and water ($T_1$ = 2607 ms). These media provided a series of $T_1$ relaxation times that entirely covered the $T_1$ range typically observed in biological tissues. Control $T_1$ measurements were performed using the inversion-recovery method with an FSE readout sequence and three-parameter fit of the corresponding signal equation (21). Transmitter gain was manually calibrated as precisely as possible. A series of scans were obtained from the phantom using the AFI sequence with $TR_1/TR_2$ = 30/150 ms (i.e., $n$ = 5) and TE = 5.6 ms at variable nominal FAs ($\alpha_{nom}$) in a range of 5–90°. The behavior of signals $S_1$ and $S_2$, and their ratio $r$ were compared with theoretical predictions.

In the second phantom experiment, the AFI method was used to correct a $T_1$ map obtained in the presence of a strongly nonuniform RF field. The experimental setup included a large uniform phantom filled with an approximately 10-mM solution of copper sulfate ($T_1$ = 138 ms) and a custom-designed surface transmit-receive coil. The phantom had a cylindrical shape and the coil was attached to its flat side. The coil consisted of a single circular loop with an approximate diameter of 8 cm and three capacitors equidistantly distributed in gaps along the circumference. $T_1$ measurements were performed using the variable-FA method (5,22). Data were acquired using the manufacturer's 3D spoiled gradient recovery (SPGR) sequence with

TR/TE = 30/3.4 ms and a set of $\alpha_{nom}$ = 5, 10, 20, 30, 40, and 60°. For $B_1$ correction, the 3D AFI sequence was applied with $\alpha_{nom}$ = 60°, $TR_1/TR_2$ = 20/100 ms ($n$ = 5), and TE = 3.4 ms. All data sets were obtained with the same geometry entirely covering the excitation region of the coil. Reconstruction of parametric images was performed using in-house-developed software written in C-language. $T_1$ maps were reconstructed by a voxel-based fit of the signal equation for the SPGR sequence:

$$S_\alpha = c\text{PD}\frac{1-\exp(-\text{TR}/T_1)}{1-\cos(c\alpha_{\text{nom}})\exp(-\text{TR}/T_1)}\sin(c\alpha_{\text{nom}}), \quad [7]$$

where $S_\alpha$ is the signal intensity at the FA $\alpha_{nom}$, $c$ is the spatially dependent $B_1$ correction factor, and the fitted parameters are $T_1$ and the effective proton density (PD). The parameter PD = $gM_0\exp(-\text{TE}/T_2^*)$ absorbs the actual spin density ($M_0$), the effect of $T_2^*$ decay, and the constant $g$ depending on the receiver gain and coil sensitivity. The correction factor $c$ reflects the relative strength of the active component of the $B_1$ field ($B_1^+$) and can be calculated as the actual/nominal FA ratio in each voxel based on the actual FA map reconstructed from the AFI data (Eq. [6]). For uncorrected reconstruction, $c$ = 1. Note that if the same coil is used for transmission and reception, the coil sensitivity determined by the counter-rotating $B_1$ field component $B_1^-$ can also be scaled by the factor $c$, assuming an identical distribution for both circularly polarized components ($B_1^+$ and $B_1^-$) (23). However, this assumption may not hold for conductive samples at high magnetic field strengths (23,24). From a practical standpoint, this circumstance may limit the applicability of the AFI correction to PD quantitation in high magnetic fields, or require additional assumptions about symmetry of $B_1^+$ and $B_1^-$ distributions in the object (24,25). At the same time, neither $T_1$ nor the actual FA measurements are affected by the $B_1^-$ component, since any possible errors in the coil sensitivity are either absorbed by the parameter PD in Eq. [7] or canceled by taking the signal ratio $r$ (Eq. [4]).

The third experiment was designed to test the sensitivity of the AFI sequence to the off-resonance effect, which potentially can be a critical source of errors in FA measurements. This experiment also aimed to verify the applicability of different RF pulse types in the AFI method. A small tube (approximately 2.5 cm in diameter) with a 1-mM solution of gadolinium ($T_1$ = 202 ms) was positioned in the center of the magnet and scanned using the transmit-receive head coil. A series of measurements were taken with $TR_1/TR_2$/TE = 8.6/43/1.9 ms ($n$ = 5) and $\alpha_{nom}$ = 60° while the scanner's center frequency was varied manually. The off-resonance profiles were measured for both pulse types described above (nonselective rectangular and selective SLR). For scans with the selective SLR pulse, the slab-selection gradient was turned off.

The AFI method was tested in vivo on two healthy male volunteers (32 and 40 years old, respectively). Informed consent was obtained from both subjects in accordance with guidelines of the institutional review board. For the first subject, a whole-body $B_1$ map was acquired with the body quadrature transmit-receive coil and the following parameters of the 3D AFI sequence: $TR_1/TR_2$/TE = 20/100/2.8 ms ($n$ = 5), $\alpha_{nom}$ = 60°, FOV = 48 × 48 cm, slice thickness = 10 mm, 3D matrix = 128 × 128 × 28, voxel size = 3.8 × 3.8 × 10.0 mm$^3$ (interpolated to 1.9 × 1.9 × 5.0 mm$^3$ after zero-filled Fourier transformation), and scan time = ~7 min. The in vivo 3D $B_1$ map was smoothed with a 3 × 3 pixel median filter for each slice.

For the second subject, the AFI method was tested for fast $B_1$ mapping and correction of 3D $T_1$ and PD maps of the head. The experiment was conducted with the manufacturer's quadrature transmit-receive head coil. The 3D AFI $B_1$ mapping sequence was applied in the coronal plane with whole-head coverage and the following parameters: $TR_1/TR_2$/TE = 7.4/37/2.3 ms ($n$ = 5), $\alpha_{nom}$ = 45°, FOV = 25 × 18 cm, slice thickness = 4 mm, 3D matrix = 96 × 60 × 60, voxel size = 2.6 × 3.0 × 4.0 mm$^3$ (interpolated to 1.0 × 1.0 × 2.0 mm$^3$ after zero-filled Fourier transformation), and scan time = 2 min 40 s. For $T_1$ mapping, a three-point variable-FA technique was used with the same volume coverage. Data were acquired using the 3D SPGR sequence with TR/TE = 12.7/5.2 ms, a set of $\alpha_{nom}$ = 3°, 12°, and 25°, 3D matrix = 256 × 96 × 116, voxel size = 1.0 × 1.9 × 2.0 mm (interpolated to 1.0 × 1.0 × 1.0 mm$^3$ after zero-filled Fourier transformation), and scan time = 2 min 22 s per volume. Parametric maps were reconstructed using Eq. [7] as described above. Prior to the $T_1$ data fit, the $B_1$ map was smoothed with a 3D median 8 × 8 × 8 voxel filter to avoid extra noise contribution into parametric maps.

## RESULTS

Figure 3 shows images obtained with the AFI sequence from the phantom containing media with variable $T_1$ and

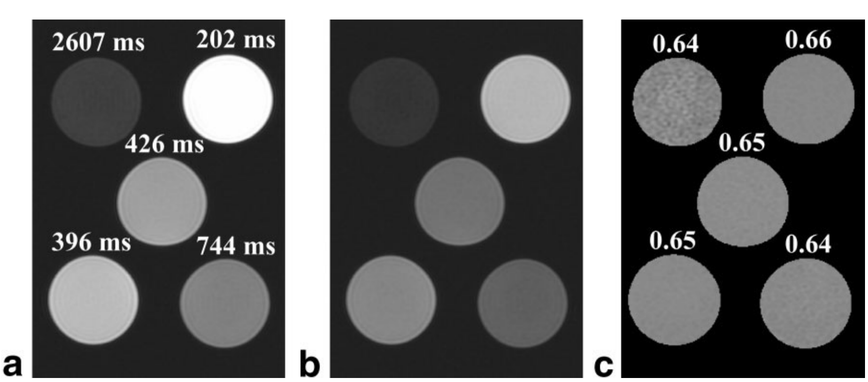


FIG. 3. Images of the phantom with variable relaxation properties obtained using the AFI sequence with $TR_1/TR_2$ = 30/150 ms ($n$ = 5) and $\alpha_{nom}$ = 60°. **a:** Image corresponding to the signal $S_1$. **b:** Image corresponding to the signal $S_2$. **c:** Calculated map of the signal ratio $r$. The numbers on images a and c indicate the $T_1$'s of the media and $r$ values, respectively. The $S_1$ and $S_2$ images are presented with identical window settings.

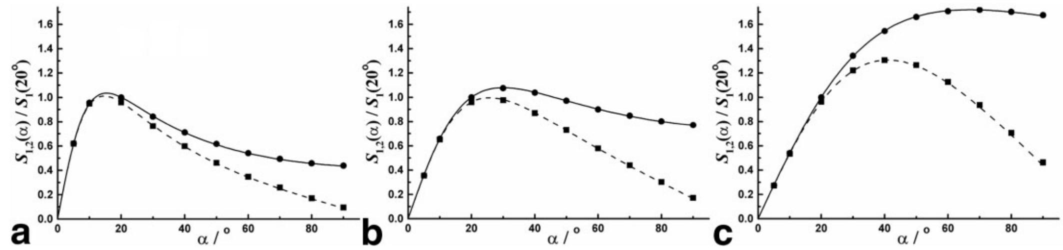


FIG. 4. Experimental (points) and theoretical (lines) intensities of the signals $S_1$ (circles and solid lines) and $S_2$ (squares and dashed lines) plotted as functions of the nominal FA. Data are presented for media with $T_1$ = 2607 ms (**a**), 744 ms (**b**), and 202 ms (**c**). Theoretical plots are simulated using Eqs. [1]–[3] for particular $T_1$ values and $TR_1/TR_2$ = 30/150 ms. For each data set, experimental and theoretical intensities are normalized to the intensity of the signal $S_1$ at the FA of 20°.

the map of the signal ratio $r$. Images corresponding to signals $S_1$ and $S_2$ demonstrate clear $T_1$-weighted contrast, while no contrast features are visible on the $r$ map. Quantitative data for this phantom are presented in Figs. 4–6. In Fig. 4, experimentally measured signal intensities at variable FAs are superimposed on simulated plots for corresponding $T_1$ values. To present data in the same scale and eliminate the unknown parameters $M_0$ and $T_2^*$ (see Eqs. [1]–[3]), both experimental and simulated intensities were normalized to the intensity of the signal $S_1$ at the FA of 20° for each sample. Figure 5 presents experimental and theoretical results for the signal ratio $r$. Both magnetization equations (Eqs. [1] and [2]) and the formula for the ratio $r$ (Eq. [4]) provide excellent agreement with experimental measurements. Experimental data for the ratio $r$ are also well described by the $T_1$-independent approximated equation (Eq. [5]), except for a minor disagreement observed at large FAs (>70°) for the sample with the shortest $T_1$ = 202 ms, which is typically beyond the physiologic range. The accuracy of FA measurements by the AFI technique is illustrated by Fig. 6, in which the measured FA values are plotted against the nominal values. Based on the initial assumptions in the experimental design, it is presumed that nominal FAs represent the actual ones due to accurate transmitter gain calibration and a negligible effect of RF nonuniformity. In the range of FA = 10–60°, the AFI data demonstrate high accuracy with errors not exceeding 1–2°. At the smallest FA of 5°, a larger dispersion of measured values is observed, presumably due to a lower SNR and a decreased sensitivity of the ratio $r$ to FA changes. Systematic errors due to the relaxation effect become visible at high FAs (>70°) as a slight underestimation of the FA. Except for the sample with very short $T_1$ = 202 ms, for other media this type of error is within 2.5° even at the largest FA of 90°. Overall, the results of this experiment appear to be in close agreement with the theoretical predictions formulated above, and confirm the applicability of the AFI method to objects with highly heterogeneous relaxation properties.

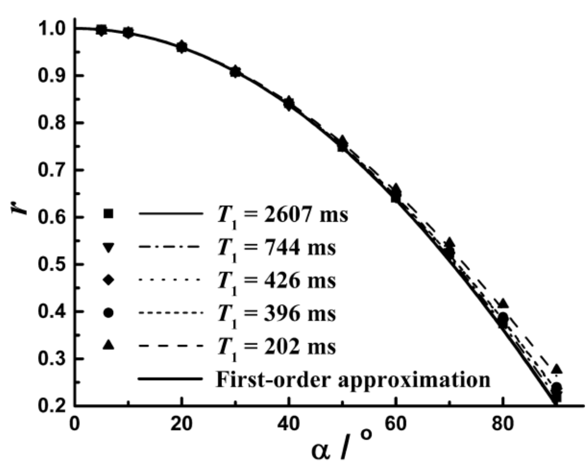


FIG. 5. Experimental (points) and theoretical (lines) signal ratios $r = S_2/S_1$ plotted as functions of the nominal FA. The thin lines represent simulations by Eq. [4] for particular $T_1$ values and $TR_1/TR_2$ = 30/150 ms. The bold solid line corresponds to the plot of the first-order approximation (Eq. [5]) at $n$ = 5. Data are presented for media with $T_1$ = 2607 ms (squares and solid line), 744 ms (down-triangles and dash-dot line), 426 ms (diamonds and dot line), 396 ms (circles and short-dash line), and 202 ms (up-triangles and dash line).

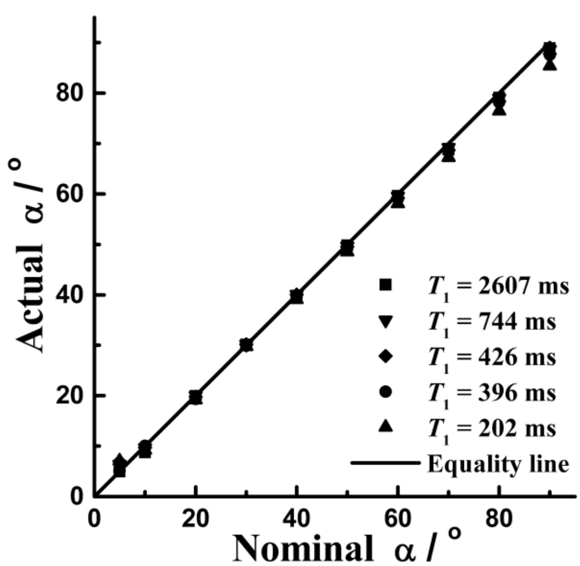


FIG. 6. FAs experimentally measured by the AFI technique at various nominal FAs in media with $T_1$ = 2607 ms (squares), 744 ms (down-triangles), 426 ms (diamonds), 396 ms (circles), and 202 ms (up-triangles). Actual FAs were calculated by Eq. [6] with $n$ = 5 from experimental ratios $r$ presented in Fig. 5. The diagonal straight line represents the ideal situation when actual and nominal FAs are equal. Note that the correlation between measured and nominal FAs for each individual sample is characterized by the correlation coefficient >0.999.

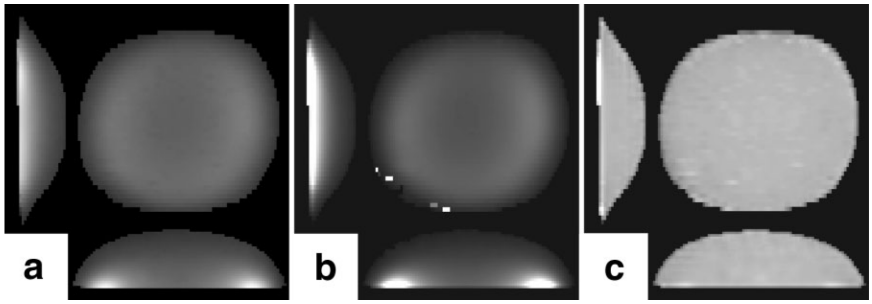


FIG. 7. Actual FA and $T_1$ mapping with the surface transmit-receive coil. **a:** Actual FA map obtained using the AFI sequence with $\alpha_{nom} = 60°$ and $TR_1/TR_2 = 20/100$ ms. The range of grayscale values represents the range of actual FAs 10–95°. **b:** Uncorrected $T_1$ map reconstructed using nominal FAs. **c:** Corrected $T_1$ map reconstructed using actual FAs. The range of grayscale values on $T_1$ maps corresponds to the range of $T_1 = 20–400$ ms. Images represent three orthogonal cross sections arbitrarily chosen from the 3D volume data sets.

Figure 7 illustrates an important practical application of the AFI method for the problem of $T_1$ measurements in the presence of strongly nonuniform RF distribution. Figure 7a presents the actual 3D FA map for the surface transmit-receive coil reconstructed from the data obtained with the AFI sequence. The FA map in Fig. 7a reflects strongly nonuniform RF excitation produced by this coil. With such a $B_1$ distribution, $T_1$ measurements based on the nominal FA values are impractical, as illustrated by the uncorrected $T_1$ map in Fig. 7b, which shows a large variety of $T_1$ values in the homogeneous object. Correcting $T_1$ by using actual FAs for each voxel results in a uniform $T_1$ map (Fig. 7c) with a narrow range of $T_1$ variations around the mean value of 143 ± 8 ms. This value is similar within the experimental error to the independently measured $T_1 = 138$ ms, while a slight overestimation likely reflects the above-discussed deviation from the first-order formalism due to the very short $T_1$ in the phantom fluid. The presented test demonstrates that $T_1$ mapping is feasible under extremely adverse experimental conditions if AFI-based data correction is applied.

Figure 8 demonstrates the off-resonance behavior of the signals $S_1$ and $S_2$ (Fig. 8a) and measured FAs (Fig. 8b) for the rectangular and minimum-phase SLR RF pulses. Excitation profiles of these pulses calculated by the numerical solution of the Bloch equations are presented in Fig. 8c for comparison. Note that the excitation profiles are symmetric relative to the zero frequency offset, and a positive frequency part is shown. An important finding is that the offset dependencies of the measured FA closely approximate excitation profiles of corresponding RF pulses. At the same time, signals $S_1$ and $S_2$ demonstrate more complicated off-resonance profiles caused by a variable degree of saturation due to frequency-dependent variations of the effective $B_1$ field. As shown in Fig. 8b, the effect of the offset frequency on the accuracy of FA measurements is minor within a relatively wide frequency range. For the rectangular pulse, the error is less than 5% in the ±0.5 kHz range. With the high-bandwidth SLR pulse, the region of reliable FA measurements exceeds ±1.5 kHz.

Figure 9 demonstrates whole-body in vivo $B_1$ mapping using the AFI method. The tissue-dependent contrast is clearly visible on the $S_1$ image (Fig. 9a) and is completely eliminated on the actual FA map (Fig. 9b), which shows only smooth intensity variations consistent with expected RF nonuniformity. Quantitatively, the variations of the actual FA span approximately from 35° (at the superior and inferior edges of the FOV) to 70° (in "hot spots" occurring in the hands and shoulders in the close proximity to the coil perimeter) across the body volume, while the values close to the nominal FA (60°) are observed in the central region of the body, as is best seen on the sagittal cross-section (Fig. 9b). It is also noticeable that the $B_1$ map produced by the AFI method is not affected by flow and

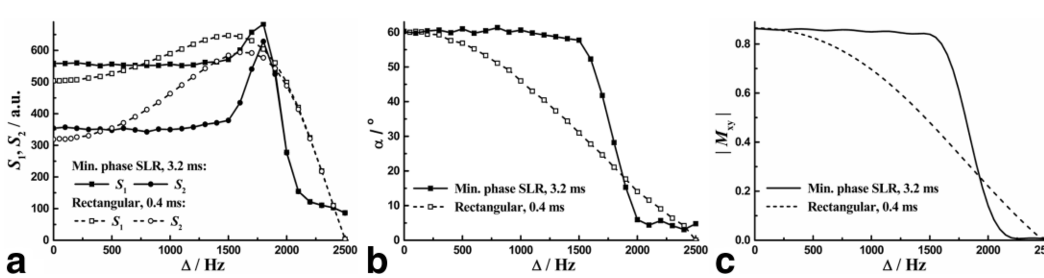


FIG. 8. Off-resonance behavior of the AFI sequence with two types of RF pulses: selective minimum-phase SLR with a 3.2-ms duration (solid symbols and solid lines) and nonselective rectangular with a 0.4-ms duration (open symbols and dashed lines). For both pulses $\alpha_{nom} = 60°$. **a:** Experimental dependence of the signals $S_1$ (squares) and $S_2$ (circles) on the offset frequency. Signal intensities are expressed in arbitrary units (a.u.). **b:** Experimental dependence of the FA calculated by Eq. [6] on the offset frequency. **c:** Theoretical excitation profiles of corresponding RF pulses computed by solving the Bloch equations. The profiles are presented as the modulus of the transverse magnetization ($M_{xy}$) after a pulse with an FA of 60°.

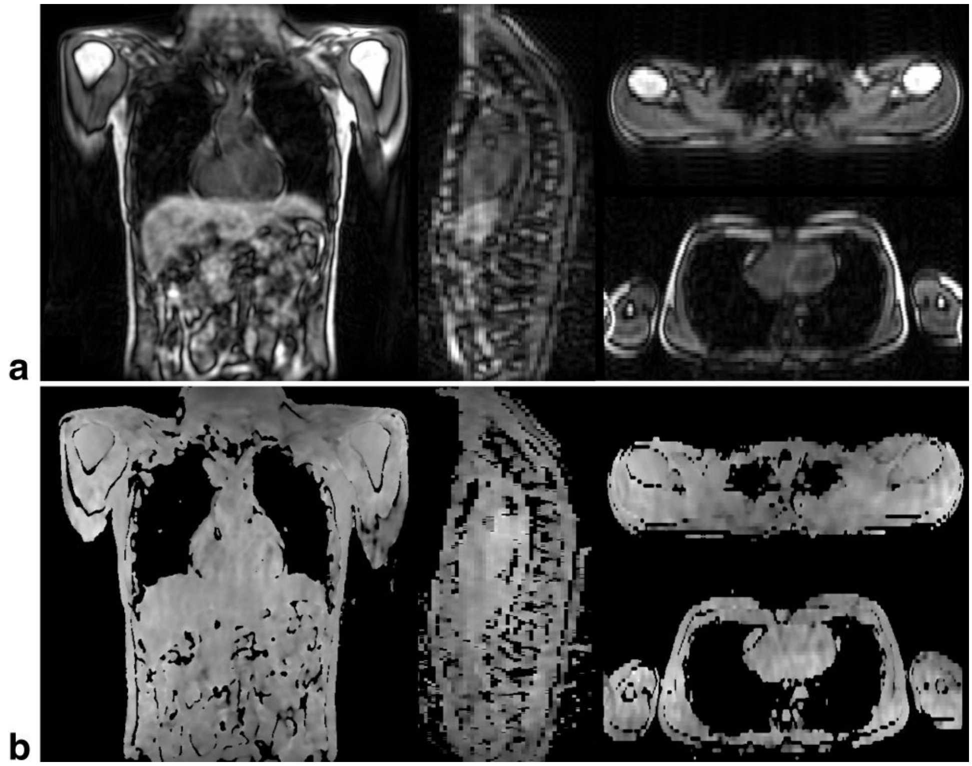

FIG. 9. In vivo $B_1$ mapping in the human body using the AFI method. **a:** Images corresponding to the signal $S_1$. **b:** Actual FA map calculated by Eq. [6]. The range of grayscale values represents the range of actual FAs 20–80°. Images correspond to four cross sections arbitrarily chosen from the 3D volume data sets.

motion artifacts, presumably because the signals $S_1$ and $S_2$ are acquired in rapid succession and have similar phase properties.

Figure 10 shows a combined application of AFI and $T_1$ mapping for the human head in vivo. The nonuniformity of the transmitted RF field produced by the transmit-receive head coil is exemplified by the $B_1$ map in Fig. 10b. As expected, $B_1$ field is highly nonuniform along the *z*-axis at the lower end of the coil. Slight nonuniformity is also visible in the *x-y* plane with a diagonal pattern and an increase of $B_1$ in the center. The range of $B_1$ variations across the brain region is from –12% to +4% relative to the nominal FA. The observed patterns and the range of $B_1$ variability are in good agreement with previous results of theoretical modeling (26–28) and experimental measurements (28–30) for a quadrature coil at 1.5T. The AFI method successfully corrects variations caused by $B_1$ inhomogeneity on both $T_1$ and PD maps (Fig. 10c–f). The effect of correction is clearly visible on sagittal

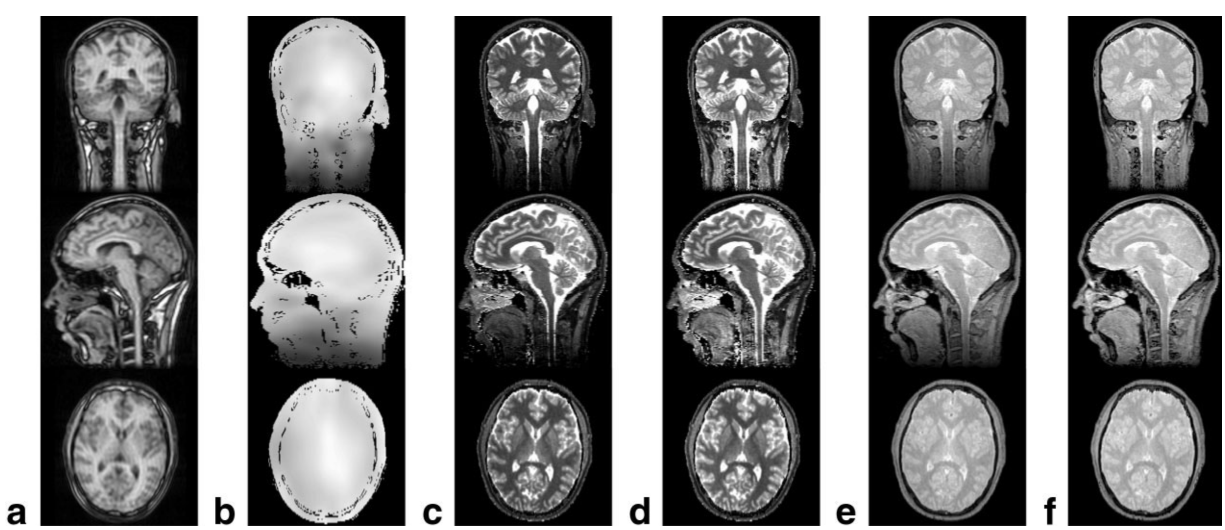

FIG. 10. Fast $B_1$ mapping and correction of $T_1$ maps in the human head in vivo. **a:** Images corresponding to the signal $S_1$. **b:** $B_1$ map calculated by Eq. [6] and normalized to the nominal FA (equivalent to the correction factor *c* in Eq. [7]). The range of grayscale values represents the 0.35–1.05 range of actual/nominal FA ratios. **c:** Uncorrected $T_1$ map. **d:** Corrected $T_1$ map. The range of grayscale values on $T_1$ maps corresponds to the range of $T_1$ = 0–2000 ms. **e:** Uncorrected PD map. **f:** Corrected PD map. All images are presented for three orthogonal cross sections arbitrarily chosen from the 3D volume data sets.

and coronal cross sections in the neck region, where the quick drop of $B_1$ results in a spurious reduction of the uncorrected parameters $T_1$ and PD (Fig. 10c and e).

## DISCUSSION

The presented method demonstrates a new principle for RF field measurements based on the pulsed steady-state formalism. In contrast to all previous methods, AFI makes use of a very short TR and does not require a substantial portion of magnetization to relax between sequence repetitions. Correspondingly, the scan time can be considerably reduced, and the method can be implemented for 3D acquisitions with large spatial coverage in a clinical setting. At the same time, the AFI technique remains highly insensitive to $T_1$ relaxation properties of the object. By definition, the AFI method is also insensitive to the effects of $T_2$, $T_2^*$, spin density, and coil reception profile. None of these factors affect the signal ratio $r$ (see Eq. [4] or [5]), since the signal readout parameters are identical for intervals $TR_1$ and $TR_2$. An important advantage of the AFI method is that images corresponding to the signals $S_1$ and $S_2$ are registered due to a very short delay between signal acquisitions. This considerably reduces the sensitivity of the technique to motion and allows application to a variety of anatomic regions. These features make AFI an almost ideal choice for $B_1$ mapping applications in vivo.

Possible limitations of the AFI method are related to the GRE nature of the acquired signals. These include the sensitivity of the signal to main magnetic field nonuniformities, chemical shift, and flow. Since $B_0$ nonuniformity and chemical shift affect both signals ($S_1$ and $S_2$) identically during readout, one can automatically eliminate them by taking the ratio $r$, unless their effects cause significant signal void or a large deviation from the resonance frequency during excitation. The flow effect may result in incorrect estimation of $B_1$ distribution since magnetization of inflowing blood may not obey the pulsed steady-state formalism used to derive signal equations (Eqs. [1] and [2]). This situation may occur if the coil coverage is insufficient to irradiate the whole volume of blood, and inflowing blood has substantially relaxed magnetization.

Based on the results of the experiments and simulations, we can formulate some basic recommendations regarding the experimental setup for AFI applications. If a large variety of $T_1$ values are expected in the object, it is safe to keep the working range of measured FAs within 20–70°. This range is sufficient to accommodate the majority of practically important $B_1$ distributions, and it allows a flexible choice of the nominal FA. The measurements can also be reliably performed for higher FAs if the presence of short relaxation components ($T_1 < 200$–$300$ ms) is not expected. Based on the expected FA range, the nominal FA should be set as high as possible to allow the maximal dynamic range of the signal ratio $r$. For example, if 10–20% $B_1$ variations are expected, it is reasonable to use the nominal FA of 55–60°. The duration of the delay $TR_1$ is typically determined by the sequence design and hardware limitations. Although this parameter does not impose strict requirements, the performance of the AFI method would benefit from using the shortest possible value. It is also worth noting that the typical timing parameters and nominal FAs in the AFI experiment result in heavily $T_1$-weighted image contrast for signals $S_1$ and $S_2$, as shown in Figs. 3, 9a, and 10a. If the total TR of the sequence is too short, the signal from tissues with long $T_1$ may appear oversaturated and thus not usable for voxel-based calculation of the ratio $r$. Figure 3 shows an example in which the low signal in the water sample with a very long $T_1$ translates into considerably increased noise on the reconstructed $r$ map. Practically, this situation must be taken into account in quantitative brain applications, where cerebrospinal fluid (CSF) may produce insufficient signal, and a reduction of the TR is motivated by the need to shorten the total scan time. One possible solution is demonstrated in the combined $B_1$ and $T_1$ mapping experiment (Fig. 10), in which the nominal FA of the AFI sequence was reduced from 60° to 45° to avoid extra saturation of the CSF signal. An alternative approach could utilize presegmentation of CSF to exclude it from image processing, since the majority of quantitative imaging applications are focused on the brain parenchyma.

Depending on a particular application, the AFI sequence can be used with non- or slab-selective RF pulses. The non-selective variant of the sequence is more suitable for coil testing and $B_1$ distribution studies in an entire object. The selective option is useful for data correction in quantitative applications targeted to a specified volume. For slab-selective applications, the spatial region for FA measurements should correspond to the flat central part of the pulse excitation profile, as shown for the selective SLR pulse in Fig. 8. RF pulses used in the AFI sequence should have a sufficiently high bandwidth to cover an expected range of frequency offsets in a particular experimental setting. For the nonselective rectangular pulse with a technically realistic duration of 0.4 ms used in this study, the AFI method tolerates a range of offset frequencies up to ~1 kHz (Fig. 8). This range translates into ~15 ppm at 1.5 T and considerably exceeds typical $B_0$ variations and chemical shifts within the human body. However, for high-field applications or in the presence of strong $B_0$ nonuniformities, pulses with a wider excitation profile may be needed. According to the pulse test results (Fig. 8), high-bandwidth selective pulses may be preferable in such situations. Based on the established similarity between the off-resonance profile of the measured FA and the pulse excitation profile (Fig. 8), the off-resonance behavior of the AFI sequence is easy to predict for any pulse configuration.

As an example of possible AFI applications, this method was used to correct results of $T_1$ mapping. Various methods for relaxation measurements are known to be prone to errors caused by RF nonuniformity and miscalibration (21). RF homogeneity is especially critical for methods that relying on exact knowledge of FA values, such as variable-FA (5,22,31) and two-point saturation recovery (7), in which errors caused by imperfect pulses cannot be deduced from the multiparameter fit (21). Several studies (5,7,31) have demonstrated the importance of performing additional $B_1$ correction procedures to obtain accurate results with these techniques. Correction of $B_1$ nonuniformity is also useful for other quantitative MR applications, such as magnetization transfer (29,30), $T_2$ mapping (12), and quantitative spectroscopic imaging (17), where the errors caused by imperfect RF pulses may introduce significant bias in parameter estimation. The present work shows that the AFI method can be successfully used in combination with the variable-FA $T_1$ mapping technique even in the situation of extremely nonuniform $B_1$ distribution (Figs. 7 and 10). Since whole-brain $B_1$ mapping with AFI takes less than 3 min, this method can be easily implemented in any quantitative imaging protocol. For protocols that do not require large anatomical coverage, one can further reduce the scan time by using slab-selective excitation targeted to a specific region of interest.

Interest in $B_1$ measurement and correction techniques has grown recently subsequent to the advent of high-field MR scanners, for which an increased RF inhomogeneity effect turns out to be a critical limitation. The present results provide an extended methodological background for the AFI method and may further facilitate its practical applications.

## CONCLUSIONS

AFI provides a new time-efficient, reliable, and simple solution for image-based measurement of the transmitted RF field. The unique feature of the proposed method is that it uses pulsed steady-state signal acquisitions. This overcomes the limitation of previous methods, which required long relaxation delays between sequence repetitions. Since it enables fast 3D measurements, AFI also avoids slice-profile problems and facilitates simple image processing. Thus the technique can easily be implemented for any in vivo application related to $B_1$ mapping or RF nonuniformity correction in quantitative methods.

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